

# Banjo Modeling

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## (a) Propagation speed of quasi-longitudinal and longitudinal waves

A thin plate can sustain different types of elastic waves, including bending, torsional, and in-plane longitudinal or quasi-longitudinal waves. Purely longitudinal waves occur only in bulk solids when the wavelength is much smaller than all structural dimensions, while in a thin plate the relevant in-plane counterpart is the quasi-longitudinal extensional wave.

For an isotropic material with Young's modulus  $E$ , density  $\rho$ , and Poisson's ratio  $\nu$ , we calculate the corresponding propagation speeds using the aluminium properties provided ( $E = 69$  GPa,  $\rho = 2700$  kg/m<sup>3</sup>,  $\nu = 0.334$ ):

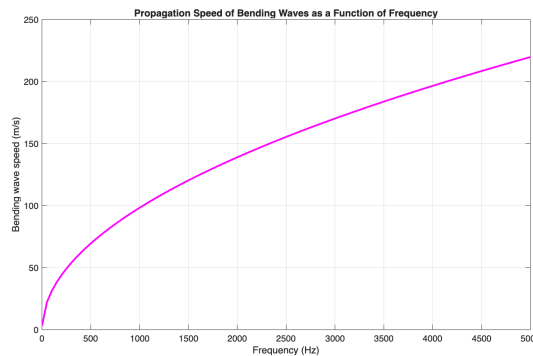
$$c_q = \sqrt{\frac{E}{\rho(1 - \nu^2)}} \approx 5.36 \times 10^3 \text{ m/s} \quad (\text{quasi-longitudinal, in-plane})$$

$$c_L = \sqrt{\frac{E(1 - \nu)}{\rho(1 + \nu)(1 - 2\nu)}} \approx 6.22 \times 10^3 \text{ m/s} \quad (\text{bulk longitudinal}).$$

## (b) Propagation speed of bending waves

In a thin plate, bending waves are dispersive: their propagation speed depends on frequency, generating distortion. For an isotropic thin plate, the bending-wave speed is:

$$c_b(f) = \sqrt{\alpha f} \quad \alpha = 1.8 h c_q$$



## (c) Modal frequencies of the first six bending modes

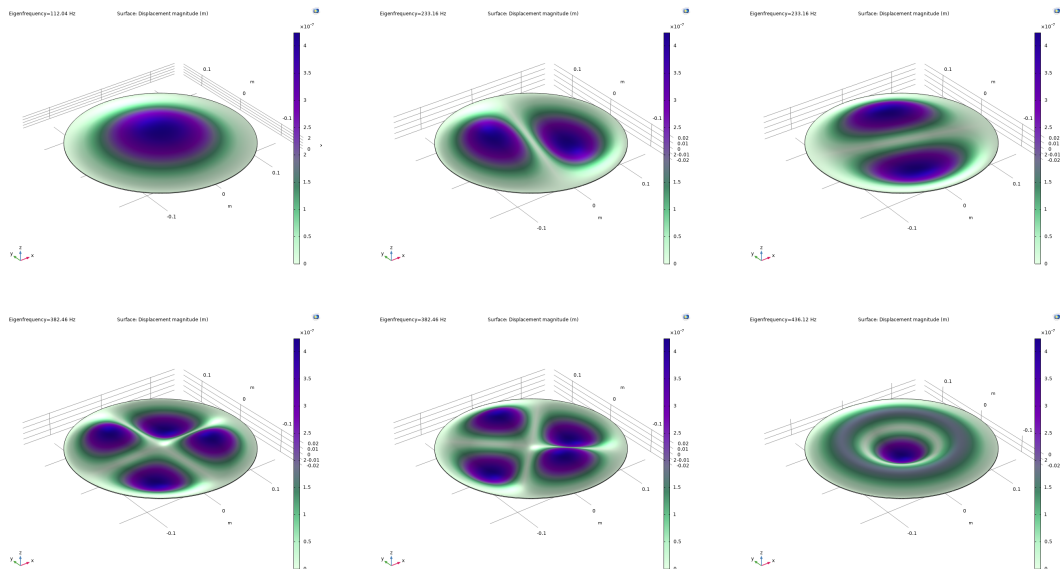
For a circular plate with clamped boundary conditions, the bending-mode frequencies can be expressed in terms of a fundamental reference frequency and a set of dimensionless modal constants (derived from known ratios for the clamped circular plate). We obtain:

$$\begin{aligned}
 f_{01} &= \frac{0.4694 c_q h}{a^2} = 111.89 \text{ Hz} & f_{11} &= 2.08 f_{01} = 232.73 \text{ Hz} \\
 f_{21} &= 3.41 f_{01} = 381.54 \text{ Hz} & f_{02} &= 3.89 f_{01} = 435.25 \text{ Hz} \\
 f_{31} &= 5.00 f_{01} = 559.45 \text{ Hz} & f_{12} &= 5.95 f_{01} = 665.74 \text{ Hz}
 \end{aligned}$$

where  $c_q$  is the quasi-longitudinal wave speed obtained in part (a),  $h = 1 \text{ mm}$  is the plate thickness, and  $a = 0.15 \text{ m}$  is the plate radius.

## (d) Eigenfrequency Analysis in COMSOL

The model was set up by using the *Solid Mechanics* interface, importing plate geometry, material properties, and boundary conditions. The first six bending modes obtained from the simulation are reported below: the eigenfrequencies are slightly higher than the predicted values, as the analytical model relies on simplified assumptions (ideal boundary conditions and thin-plate), while the finite-element simulation accounts more accurately for geometry and stiffness distribution. The deviation is small enough to confirm that the analytical formulation is consistent and the numerical model is accurate.



## (e) Point Mobility Simulation in COMSOL

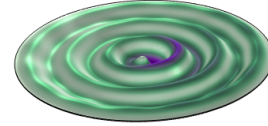
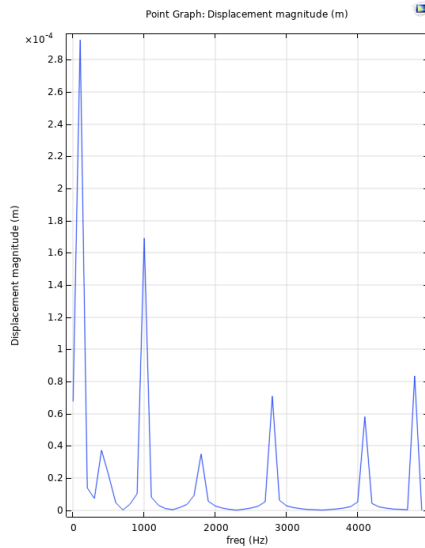
To compute the point mobility  $Y_c$  of the plate, a geometric point was created and a *Point Load* was assigned to it, applying  $F_z = 1$  N,  $F_x = F_y = 0$ .

A *Frequency Domain* study was added with `range(1, 100, 5000)`, (from 1 Hz to 5000 Hz with steps of 100 Hz), obtaining the displacement response at the excited point through a *1D Point Graph*.

$$Y_c(\omega) = \frac{U(\omega)}{F_0} = U(\omega) \quad \text{for } F_0 = 1 \text{ N.}$$

In the resulting mobility curve, pronounced peaks appear at the resonance frequencies of the plate, as expected from the theoretical modal behavior.

The deformation at 4901 Hz is reported as a representative case, illustrating the spatial response pattern of the plate under the string-like excitation.



## (f) Tuning the string to the first plate mode

To evaluate the interaction between a given string and the soundboard, we match the string fundamental frequency  $f_{s,1}$  and the first bending mode of the plate  $f_{01}$ , for an ideal (flexible, mass-loaded) string with fixed ends:

$$f_{s,1} = \frac{1}{2L} \sqrt{\frac{T}{\mu}} \quad \Rightarrow \quad T = (2Lf_{01})^2 \mu$$

$L$  string length,  $T$  tension,  $\mu = \rho A = \rho \pi r^2$  string linear mass density.

## (g) Coupled String-Plate System

We analyze the coupling between the string and the plate (bending stiffness of string neglected), which modifies the modal frequencies of the system and the vibration behavior of the two structures in two different scenarios:

$$\text{Weak coupling: } \kappa_{mn} < \frac{\pi^2}{4Q_{mn}^2} \quad \text{Strong coupling: } \kappa_{mn} \geq \frac{\pi^2}{4Q_{mn}^2}.$$

$$m_s = \rho_s A_s L = \rho_s \pi r_s^2 L \quad m_p = \rho V = \rho \pi a^2 h \quad \kappa_{mn} = \frac{m_s}{n^2 m_p}$$

For strong coupling, the coupled modal frequencies are approximately:

$$\tilde{f}_{mn} \approx f_{mn} \sqrt{1 + \frac{m_s}{m_p}}.$$

Evaluating in MATLAB the coupling parameter  $\kappa_{mn}$  for each combination of the  $n$ th mode of the string and the  $m$ th mode of the plate (total of 36 combinations), we observe that the strong coupling condition is satisfied for 10 of them. However, strong coupling is only impactful on the modal frequencies if the relative frequency difference between the string and plate modes is sufficiently small:  $-0.08 \leq \Delta_{rel} \leq 0.08$  with  $\Delta_{rel} = \frac{\omega_{Si} - \omega_{Bj}}{\omega_{Bj}}$ .

## (h) Plate Admittance

In this section, the mechanical admittance of the soundboard is computed as the sum of contributions from its individual modes:

$$Y(\omega) = \sum_{k=1}^K Y_k(\omega) \quad Y_k(\omega) = \frac{i\omega}{\omega^2 - \omega_k^2 + 2i\alpha_k\omega}$$

$\omega_k$  angular frequency of the  $k$ -th mode;  $\alpha_k = \omega_k/(2Q_k)$  modal damping. Each single-mode admittance is normalized such that:

$$\max_{\omega} |Y_k(\omega)| = 1 \text{ s/kg}.$$

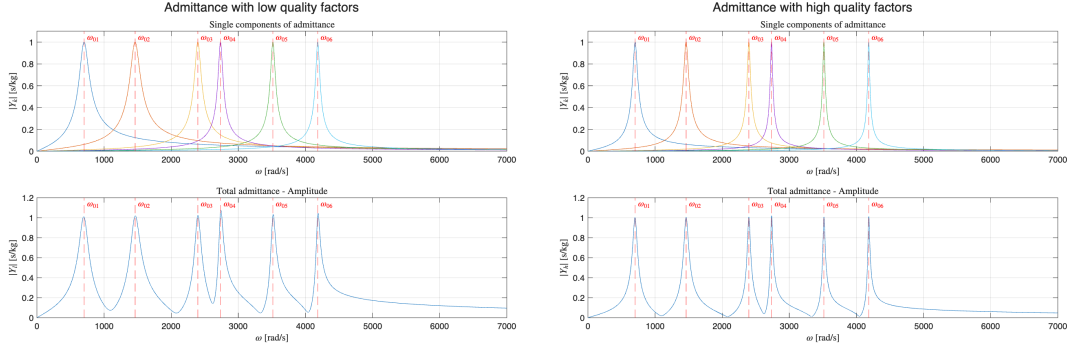
Two different sets of merit factors are considered:

Low quality factors:  $Q = [5, 10, 25, 40, 50, 70]$ , producing the admittance  $Y_l(\omega)$ .

High quality factors:  $Q = [10, 20, 50, 80, 100, 140]$ , producing  $Y_h(\omega)$ .

The total admittance is obtained by summing the normalized contributions of all modes:

$$Y_{tot}(\omega) = \sum_{k=1}^K \frac{Y_k(\omega)}{\max |Y_k(\omega)|}.$$



Increasing the merit factors gives sharper and more pronounced peaks in the admittance: lower damping and higher resonance amplitudes for each mode.

## (i) Helmholtz Resonance of the Soundbox

The Helmholtz resonance of a soundbox with circular aperture can be modeled as a lumped mass–spring system: the air in the aperture acts as the oscillating mass and the air in the cavity as the compliance. The frequency is:

$$f_H = \frac{c}{2\pi} \sqrt{\frac{A}{V L_{\text{eff}}}} \quad A = \pi a^2$$

$c$  speed of sound,  $a$  aperture radius,  $V$  cavity volume,  $L_{\text{eff}}$  effective neck length.

**Neckless aperture:** aperture modeled as a hole in a thin plate (no physical neck).  $L_{\text{eff}} \approx 1.7 a$  (approximated by the radiation end–correction), therefore:

$$f_{Hole} \approx \frac{c}{2\pi} \sqrt{\frac{\pi a^2}{V(1.7 a)}} = \frac{c}{2\pi} \sqrt{\frac{\pi}{1.7} \frac{a}{V}} \approx 81.014 \text{ Hz.}$$

**Aperture with neck:** aperture regarded as a short cylindrical neck of length  $L$  (for instance equal to the plate thickness).  $L_{\text{eff}} = L + 1.7 a$ , therefore:

$$f_{Neck} = \frac{c}{2\pi} \sqrt{\frac{\pi a^2}{V(L + 1.7 a)}} \approx 746.508 \text{ Hz.}$$

## (j) Admittance with Helmholtz Mode considered

The dynamic response of the plate-soundbox system is analyzed by including the Helmholtz resonance of the cavity. This mode arises from the compressibility of the air in the soundbox and the mass of air in the aperture, and can be modeled as a single-degree-of-freedom harmonic oscillator.

The angular frequency of the Helmholtz mode,  $\omega_H$ , is given by:

$$\omega_H = 2\pi f_{\text{Helm}} = 2\pi \frac{c}{2\pi} \sqrt{\frac{1.85 A_{\text{ap}}}{V}},$$

where  $c$  is the speed of sound,  $V$  is the cavity volume, and  $A_{\text{ap}}$  is the aperture area. Its admittance can be expressed as

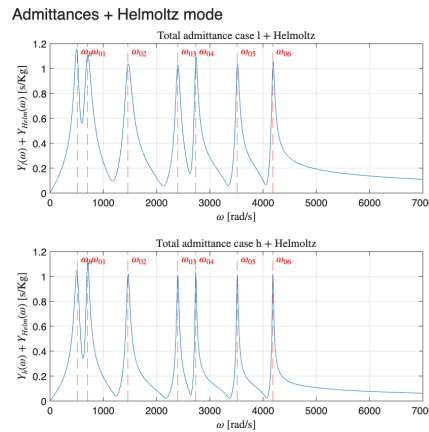
$$Y_{\text{Helm}}(\omega) = \frac{i\omega}{\omega^2 - \omega_H^2 + 2i\alpha_H\omega},$$

with damping  $\alpha_H = \omega_H/(2Q_H)$  and  $Q_H = 5$ . The Helmholtz admittance is normalized so that its maximum magnitude is unity.

The total system admittance is obtained by summing the plate admittance with the normalized Helmholtz contribution:

$$Y_{\text{total}}(\omega) = Y_{\text{plate}}(\omega) + Y_{\text{Helm}}(\omega),$$

yielding  $Y'_l(\omega)$  and  $Y'_h(\omega)$  for the low- and high-quality factor cases, respectively. Plots of these admittances show the Helmholtz mode as a low-frequency peak superimposed on the plate's modal contributions, representing the coupled behavior of the soundboard and resonator.



## (k) Synthesis of the Plucked String Sound

The sound of the string is synthesized by modeling its vibration when plucked at one-fifth of its length. In this approach, the string is considered ideal, neglecting the coupling with the plate. The initial displacement profile of the string can be represented as a triangular shape, which can be expanded into a Fourier series:

$$y(x, 0) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right), \quad B_n = \frac{2}{L} \int_0^L y(x, 0) \sin\left(\frac{n\pi x}{L}\right) dx.$$

The time evolution of each harmonic is given by

$$y_n(t) = B_n \cos(\omega_n t) e^{-t/\tau},$$

where  $\omega_n = n\pi\sqrt{T/\mu}/L$  is the angular frequency of the  $n$ -th mode,  $T$  is the string tension,  $\mu$  its linear mass density, and  $\tau = 1.3$  s is the decay time of the plucked string.

To account for the interaction with the resonator, the previously computed admittances  $Y_l'(\omega)$  and  $Y_h'(\omega)$  of the plate-soundbox system, together with the string admittance  $Y_c(\omega)$ , are used as linear filters applied to each harmonic component. The filtered output for each case is obtained by multiplying the Fourier amplitude of each mode by the magnitude of the corresponding admittance and adding the phase shift:

$$y_{n,\text{out}}(t) = B_n |Y(\omega_n)| \cos(\omega_n t + \arg Y(\omega_n)) e^{-t/\tau}.$$

The total synthesized signal is then obtained by summing all harmonics. Separate outputs are produced for the low- and high-quality factor admittance cases, denoted  $O_l(t)$  and  $O_h(t)$ , which represent the string filtered by the coupled system including the Helmholtz resonance.

